

HAT-P-36b

R-1528

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200528 · created 2026-08-01T17:51:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458997.69451 +/- 0.00056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.01
Transit depth (Rp/Rs)^2	2.48 +/- 0.32 [%]
Orbital Inclination (inc)	81.23 +/- 0.86
Airmass coefficient 1 (a1)	0.988 +/- 0.01
Airmass coefficient 2 (a2)	0.012 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	#2 - [436, 209]
Optimal Aperture	3.71
Optimal Annulus	12.82
Transit Duration (day)	0.0772 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.01 vs 0.126 (deviation 1.99σ)
Duration fit vs published	0.0772 d vs 0.0925 d (deviation 3.93σ)
Mid-time O–C	-9.0 min
Depth SNR	10.1
Residual scatter	1.03 %

Light curve

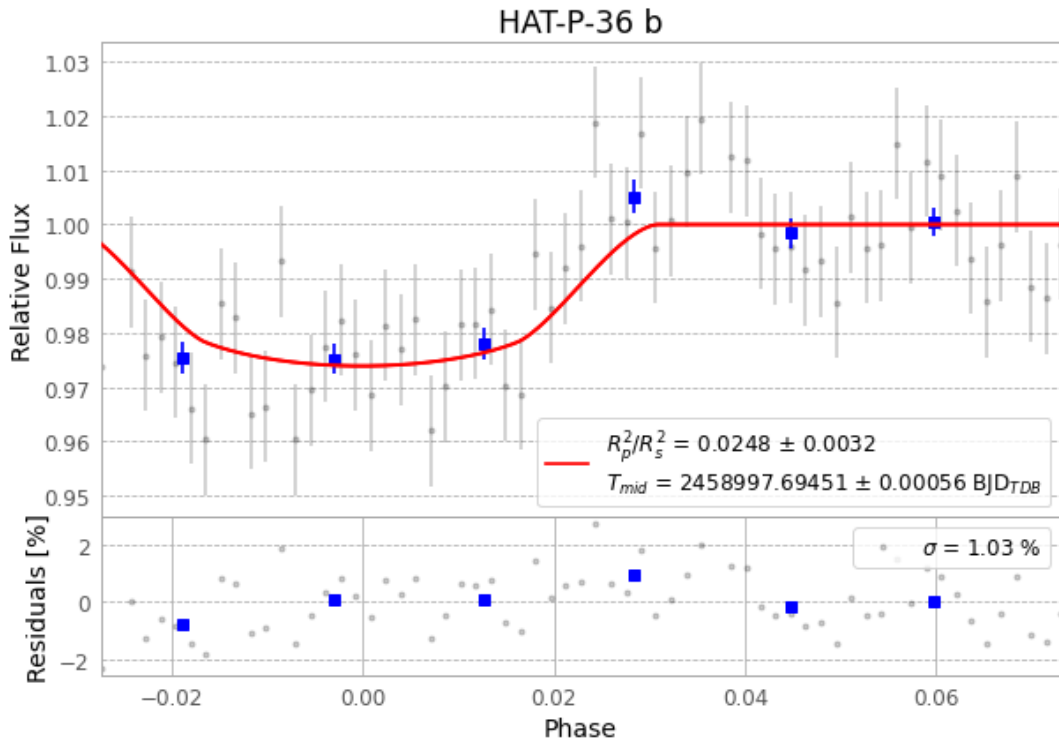


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-05-27.png (SHA-256 f56566a4e1198ac2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b02ff2c127c3eba5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), HATP-36200528034513.FITS → HATP-36200528065713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200528.log (cbb16e85753e...), FinalLightCurve_HAT-P-36 b_2020-05-27.png (f56566a4e119...), FinalLightCurve_HAT-P-36 b_2020-05-27.csv (45216a4d9842...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:51Z.